

SUJATA BASNET

Civil Engineer | Utah State University | Old Main Hill, Logan, Utah | 84322
Email: sujatabasnet216557@gmail.com | +1-435-512-2449 | <https://www.linkedin.com/in/sujata-basnet-3506ab20b/>

CAREER OBJECTIVE

To leverage my advanced education in water resources & hydrologic modeling, combined with my research and analytical skills, to contribute to a sustainable future for our water infrastructures & climate change

RESEARCH INTEREST

Integrated Water Resource Management; Ground-water; Hydrologic & Flood Modeling; Big Data Analysis & Machine Learning; Climate Change Impact on Water Resources; Hydroinformatics

EDUCATION

Ph.D. in Engineering Education, Utah State University, Logan, Utah (GPA :3.9) **Jan 2024-Present**

- Ongoing Dissertation: “Psychometric Evaluation and Cognitive Correlates of the Tactile Mental Cutting Test: Investigating Reliability, Validity, and Gender Differences in Spatial Ability, Executive Function and Social Emotional Competence among Native American Elementary Students”

Master of Engineering (Water Engineering and Management) (GPA: 3.72) **Aug 2021 -May 2023**

Asian Institute of Technology (AIT), Thailand

- Master Thesis: “Ecosystem-Based Adaptations for Groundwater Resources Management in Siem Reap Province, Cambodia”

Bachelor of Engineering: Civil Engineering **Oct 2012-Sep 2016**

Advanced College of Engineering and Management (ACEM), Lalitpur, Nepal

- Research Project: Small-scale water supply system in a rural part of Nepal

TECHNICAL & PROFESSIONAL SKILLS

- Numerical modeling (hydrologic and groundwater), including WetSpa, GMS-MODFLOW, PCSWMM, CROPWAT, and Irripro
- Knowledge in the use of CAD, GIS applications (ArcGIS, QGIS)
- Experience in geospatial programming using Python, R.
- Experience in Data Analysis using Python
- Prepared engineering reports, specifications, datasheets, and AutoCAD design drawing
- Microsoft Office Packages (Word, Excel, PowerPoint, Outlook)
- Municipal engineering work requiring the application of Civil Engineering Principles and Practices
- Strong verbal and written communication abilities, self-drive, teamwork practices, data analysis, technical

problem solving, and capacity to multitask well in a fast-paced environment

RESEARCH EXPERIENCE

Graduate Research Assistant(Jan 2024-Present):PDG Community

(May 2022 – May 2023): Research Student, Ecosystem-Based Adaptation for Sustainable Groundwater Resources Management in the Transboundary Cambodia-Vietnam Mekong Delta Aquifer, Lower Mekong Region (*Funded by SEI*)

- Developed a Groundwater Model of Siem Reap, Cambodia.
- Conducted a Social survey in two different districts of Siem Reap
- Collaborated with the Institute of Technology of Cambodia (ITC)

PROFESSIONAL EXPERIENCE

9th Feb-31st Aug 2022: Project Assistant, Asian Institute of Technology, Thailand

- Preparation of the teaching and training materials
- Participation in projects related to the subject materials.

May 2018-Present: **Non-gazetted Technical Officer, Suryabinayak Municipality**, Office of Municipal Executive, Bhaktapur, Nepal

- Analysis, Assessment, Planning, Designing, Estimating, and implementation of various infrastructure development works like Municipal Water Supply System and Sewerage Projects
- Flood Modelling of Hanumante River, Bhaktapur Municipality, Nepal
- Shallow tube well feasibility survey and design of water supply system in various communities of Suryabinayak Municipality

Jan 2018- May 2018: **Non-gazetted Technical Officer, District Soil and Conservation Office**, Ramechhap Nepal

- Estimation, procurement, and implementation supervision of various infrastructure River Training and Soil Conservation

Jul 2017 – Jan 2018: **Engineer, Central Level Project Implementation Unit**, Kathmandu Nepal

- SAP modeling, CAD Design, Estimating, and supervision of various Earthquake Resistant Building Projects
- Site Supervision and Monitoring of various Governmental Buildings

CONFERENCE PRESENTATIONS & ATTENDANCE

- **22-25th June 2025:** 2025 ASEE Annual Conference & Exposition-**Presenter**,*Work in Progress: Exploring Reliability of the Tactile Mental Cutting Test in Assessing Spatial Abilities Among Native American Children*
- **12th July 2023:** *E-learning course on Nature-based Solutions for Disaster and Climate Resilience*

- **2nd March 2023:** International Conference on Water Management Modeling-**Presenter**, *Ecosystem-Based Adaptation for Sustainable Groundwater Resources Management in the Transboundary Cambodia-Vietnam Mekong Delta Aquifer, Lower Mekong Region*
- **21st Dec 2022:** *E-learning course on Ecosystem Based Adaptation: Working with nature to adapt to a changing climate*
- **9-11th Nov 2022:** SUMERNET Annual Meeting, *Ecosystem-Based Adaptation for Sustainable Groundwater Resources Management in Transboundary Cambodia-Mekong River Delta Aquifer, Lower Mekong region*
- **29th Sep 2021:** Second Regional Workshop on *Mapping groundwater resilience to climate change and human development in Asian Cities*
- **22nd Sep 2021:** AIT-BNU Virtual Workshop on *Climate change, water resources, and Ecosystem; Challenges and Opportunities*
- **9th Sep 2021:** Webinar on *Nature Based Solutions for Climate change adaptation: From Concept to Implementation*

AWARDS

- Scholarship for Ph.D.Studies
- Nominated for Best Master Thesis Award under The Prof. Emeritus Madan Mohan Das Memorial.
- AIT Scholarship for Graduate Studies

PROFESSIONAL MEMBERSHIP

- Member of American Society of Engineering Education
- Member of Nepal Engineering Council (NEC) Reg. No 15258 Civil “A” Category
- General Member of Nepal Engineering Association (NEA)

PUBLICATION

- Basnet, S., Goodridge, W. H., & Kane, D. (2025, June). Work in Progress: Exploring Reliability of the Tactile Mental Cutting Test in Assessing Spatial Abilities Among Native American Children. In *2025 ASEE Annual Conference & Exposition*. <https://peer.asee.org/work-in-progress-exploring-reliability-of-the-tactile-mental-cutting-test-in-assessing-spatial-abilities-among-native-american-children>